

TP1 – 8-bit Binary Counter (0 to 255) and Modulo-10 Display on a Common-Cathode 7-Segment Display

1. Objective

Develop a LabVIEW VI that: (i) counts from 0 to 255 and displays the value in decimal and binary; (ii) generates a modulo-10 digit and sends it to a common-cathode 7-segment display.

2. Theoretical Background

- An 8-bit counter encodes 256 states: from 00000000 to 11111111, corresponding to 0 to 255.
- The i terminal of the While Loop generates an increasing integer sequence.
- Using Quotient & Remainder with divisor 256, the Remainder output creates a cyclic counter from 0 to 255.
- Using divisor 10, the Remainder output creates a modulo-10 counter for digits 0 to 9.
- For a common-cathode 7-segment display, a segment turns on when its logic state is 1.

3. Required Equipment

NI ELVIS II+, LabVIEW, NI ELVISmx, breadboard, one common-cathode 7-segment display, eight 220-ohm resistors, and connecting wires.

4. Expected Results

The main counter must progress as 0, 1, 2, ..., 255 and then return to 0. The Front Panel must display the decimal and binary values. The 7-segment display must show 0, 1, 2, ..., 9 and then repeat.

5. Discussion / Practical Questions

- Implement in LabVIEW an 8-bit binary counter ranging from 0 to 255.
- Explain why an 8-bit code can represent 256 values, from 0 to 255.
- Experimentally verify that the counter correctly displays 00000000 for 0 and 11111111 for 255.
- Test and explain the automatic return to 0 after the value 255.
- State the role of the Quotient & Remainder block in implementing the binary counter.
- Implement a modulo-10 counter that generates the sequence 0, 1, 2, ..., 9, then returns to 0.
- Establish the correspondence table between digits 0 to 9 and segments a, b, c, d, e, f, g of a common-cathode 7-segment display.
- Experimentally verify that each digit shown on the 7-segment display matches the value sent by the program.
- Identify the possible causes of incorrect display on the 7-segment device: code error, wiring error, segment inversion, or incorrect DIO line selection.
- Explain the effect of the milliseconds to wait parameter on the counting speed in both parts of the lab.

6. Example of the obtained interface

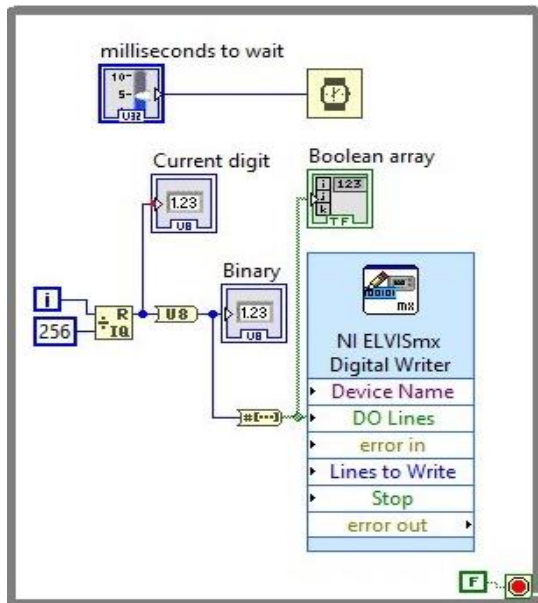


Figure 1 : Block Diagram of the 8-bit Binary Counter

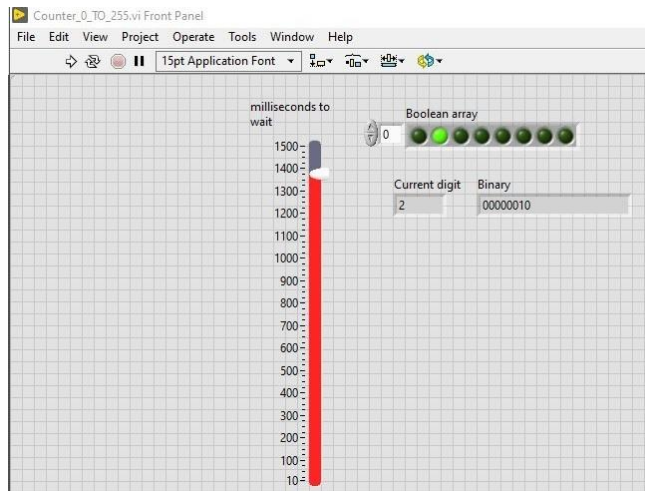


Figure 2 : Front Panel of the 8-bit Binary Counter

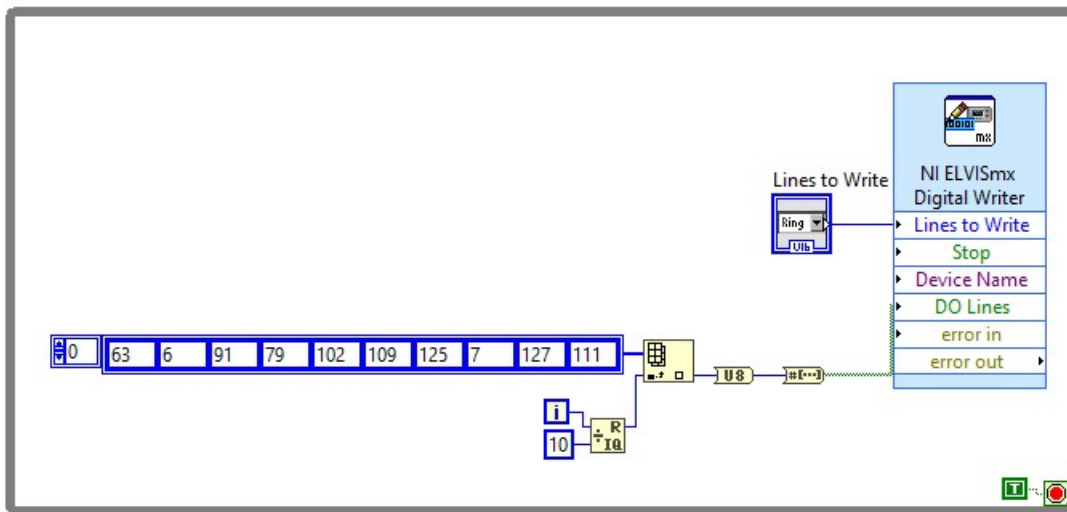


Figure 3 : Block Diagram of the Modulo-10 Display on a 7-Segment Display

Appendix – Pin Mapping Index and Coding Table

Adopted convention: bit0 = a, bit1 = b, bit2 = c, bit3 = d, bit4 = e, bit5 = f, bit6 = g, and bit7 = dp. The decimal point dp remains set to 0 if it is not used.

Bit	ELVIS	Segment
b0	DIO0	a
b1	DIO1	b
b2	DIO2	c
b3	DIO3	d
b4	DIO4	e
b5	DIO5	f
b6	DIO6	g
b7	DIO7	dp

Note: connect both common-cathode pins to ground (GND). The physical pin numbering may vary depending on the display model, so the datasheet must be checked before wiring.

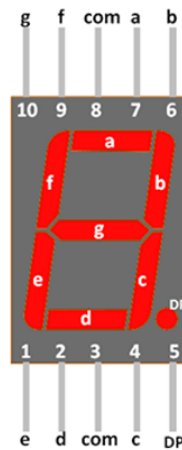


Figure 4 : 7 Segment Display Pinout